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1.2 Fractions, Decimals and Percentages

OCR GCSE Mathematics (J560) — Component 01 (Paper 1, Non-Calculator), Component 02 (Paper 2, Calculator), Component 03 (Paper 3, Calculator)

Quick Reference: FDP Conversion Table

Fraction	Decimal	Percentage	Fraction	Decimal
1/2	0.5	50%	1/8	0.125
1/4	0.25	25%	1/3	0.333...
3/4	0.75	75%	2/3	0.666...
1/5	0.2	20%	1/10	0.1

Common fraction-decimal-percentage equivalences. These should be memorised for Paper 1 (non-calculator).

1.2.1 Types of Fractions and Equivalence

- **Proper fraction:** numerator < denominator, e.g. $\frac{3}{5}$. Value is less than 1.
- **Improper fraction:** numerator ≥ denominator, e.g. $\frac{7}{4}$. Value is 1 or more.
- **Mixed number:** whole number + proper fraction, e.g. $1\frac{3}{4}$.
- **Equivalent fractions:** fractions with the same value, e.g. $\frac{1}{2} = \frac{3}{6} = \frac{50}{100}$.

Converting Between Improper Fractions and Mixed Numbers

Worked Example:

Convert $\frac{17}{5}$ to a mixed number.

Divide: $17 \div 5 = 3$ remainder 2

$$\frac{17}{5} = 3\frac{2}{5}$$

Worked Example:

Convert $2\frac{3}{7}$ to an improper fraction.

Multiply whole number by denominator, add numerator: $2 \times 7 + 3 = 17$

$$2\frac{3}{7} = \frac{17}{7}$$

Simplifying Fractions

Divide numerator and denominator by their HCF. A fraction is in its **simplest form** (or lowest terms) when the only common factor of top and bottom is 1.

Worked Example:

Simplify $\frac{36}{48}$.

$\mathrm{HCF}(36, 48) = 12$

$$\frac{36}{48} = \frac{36 \div 12}{48 \div 12} = \frac{3}{4}$$

Equivalent Fractions

Multiply or divide both numerator and denominator by the same number.

$$\frac{2}{3} = \frac{2 \times 5}{3 \times 5} = \frac{10}{15}$$

Exam Tip: Always simplify your final fraction answer unless the question asks you to leave it in a specific form. Marks can be lost for unsimplified answers.

1.2.2 Operations with Fractions

Adding and Subtracting Fractions

Fractions must have the same denominator (a **common denominator**) before they can be added or subtracted. Find the LCM of the denominators.

Worked Example:
Calculate $\frac{3}{4} + \frac{5}{6}$

LCM(4, 6) = 12

$$\frac{3}{4} + \frac{5}{6} = \frac{9}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$$

Worked Example:
Calculate $3\frac{1}{2} - 1\frac{2}{3}$

Convert to improper fractions: $\frac{7}{2} - \frac{5}{3}$

LCM(2, 3) = 6

$$\frac{7}{2} - \frac{5}{3} = \frac{21}{6} - \frac{10}{6} = \frac{11}{6} = 1\frac{5}{6}$$

Multiplying Fractions

Multiply numerators together and denominators together. Cancel common factors first to keep numbers small (this is called **cross-cancelling**).

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Worked Example:
Calculate $\frac{4}{9} \times \frac{3}{8}$

Cross-cancel: 4 and 8 share factor 4; 3 and 9 share factor 3

$$\frac{\cancel{4}^{1}}{9_{\cancel{3}}} \times \frac{\cancel{3}^{1}}{\cancel{8}^{2}} = \frac{1 \times 1}{3 \times 2} = \frac{1}{6}$$

Worked Example:

Calculate $1\frac{2}{3} \times 2\frac{1}{4}$

Convert: $\frac{5}{3} \times \frac{9}{4}$

Cancel: 9 and 3 share factor 3

$$\frac{5}{\cancel{3}} \times \frac{\cancel{9}^{3}}{4} = \frac{5 \times 3}{1 \times 4} = \frac{15}{4} = 3\frac{3}{4}$$

Dividing Fractions

To divide by a fraction, **flip the second fraction** (find its reciprocal) and multiply. Remember: *Keep, Change, Flip* (KCF).

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Worked Example:

Calculate $\frac{5}{6} \div \frac{2}{3}$

$$\frac{5}{6} \div \frac{2}{3} = \frac{5}{6} \times \frac{3}{2} = \frac{15}{12} = \frac{5}{4} = 1\frac{1}{4}$$

Worked Example:

Calculate $2\frac{1}{2} \div 1\frac{1}{4}$

Convert: $\frac{5}{2} \div \frac{5}{4}$

$$\frac{5}{2} \times \frac{4}{5} = \frac{\cancel{5}}{\cancel{2}} \times \frac{\cancel{4}^{2}}{\cancel{5}} = \frac{1 \times 2}{1 \times 1} = 2$$

Exam Tip: Always convert mixed numbers to improper fractions BEFORE multiplying or dividing. Trying to work with mixed numbers directly leads to errors.

1.2.3 Converting Between FDP

Fraction to Decimal

Divide the numerator by the denominator.

$$\frac{3}{8} = 3 \div 8 = 0.375$$

Decimal to Fraction

Write the decimal over the appropriate power of 10, then simplify.

$$0.45 = \frac{45}{100} = \frac{9}{20}$$

Fraction/Decimal to Percentage

Multiply by 100.

$$\frac{7}{20} = 0.35 = 35\%$$

$$0.625 \times 100 = 62.5\%$$

Percentage to Fraction/Decimal

Divide by 100.

$$37\% = \frac{37}{100} = 0.37$$

Recurring Decimals

A **recurring decimal** is one where digits repeat forever. Notation: a dot above the repeating digit(s).
$0.\dot{3} = 0.333...$ $0.\dot{1}\dot{8} = 0.181818...$ $0.1\dot{6} = 0.1666...$

To convert a recurring decimal to a fraction, use the algebraic method:

Worked Example:
Convert $0.\dot{3}$ to a fraction.

Let $x = 0.3333...$

Then $10x = 3.3333...$

Subtract: $10x - x = 3.3333... - 0.3333... = 3$

$9x = 3 \Rightarrow x = \frac{3}{9} = \frac{1}{3}$

Worked Example:
Convert $0.\dot{1}\dot{8}$ to a fraction.

Let $x = 0.181818...$

Then $100x = 18.181818...$

Subtract: $99x = 18 \Rightarrow x = \frac{18}{99} = \frac{2}{11}$

Worked Example:
Convert $0.1\dot{6}$ to a fraction.

Let $x = 0.1666...$

$10x = 1.666...$ and $100x = 16.666...$

$100x - 10x = 16.666... - 1.666... = 15$

$90x = 15 \Rightarrow x = \frac{15}{90} = \frac{1}{6}$

Exam Tip: Multiply by 10^n where n is the number of digits in the repeating block. If one digit repeats, multiply by 10; if two digits repeat, multiply by 100; and so on.

1.2.4 Percentage Calculations

Finding a Percentage of an Amount

Multiply the amount by the percentage expressed as a decimal (the **multiplier**).

$$\text{Amount} \times \frac{\text{percentage}}{100}$$

Worked Example:
Find 35% of £240.

$$240 \times 0.35 = £84$$

Or: 10% of 240 = 24; 30% = 72; 5% = 12; 35% = 72 + 12 = £84

Percentage Increase and Decrease

- **Multiplier for an increase:** $1 + \frac{r}{100}$ e.g. 15% increase: multiplier = 1.15
- **Multiplier for a decrease:** $1 - \frac{r}{100}$ e.g. 20% decrease: multiplier = 0.80

Worked Example:
A jacket costs £85. It is increased by 12%. Find the new price.

$$85 \times 1.12 = £95.20$$

Worked Example:

A car costs £14,000. After a 15% discount, find the sale price.

$$14{,}000 \times 0.85 = £11{,}900$$

Percentage Change

$$\text{Percentage change} = \frac{\text{change}}{\text{original}} \times 100$$

Worked Example:

A share was worth £2.40 and rose to £2.64. Find the percentage increase.

$$\text{Change} = 2.64 - 2.40 = 0.24$$

$$\text{Percentage increase} = \frac{0.24}{2.40} \times 100 = 10\%$$

Reverse Percentages (Finding the Original Amount)

When you know the value *after* a percentage change, divide by the multiplier to find the original.

Worked Example:

After a 20% increase, a price is £96. Find the original price.

Multiplier = 1.20 (for a 20% increase)

$$\text{Original} = 96 \div 1.20 = £80$$

Worked Example:

A TV costs £374 after a 15% reduction. What was the original price?

Multiplier = 0.85 (for a 15% decrease)

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$$\text{Original} = 374 \div 0.85 = £440$$

Exam Tip: Never calculate a reverse percentage by adding or subtracting the percentage from the sale price. For example, if a price was reduced by 20% to give £80, the original is NOT £80 + 20% of £80 = £96. The correct method is £80 ÷ 0.80 = £100.

1.2.5 Compound Interest and Depreciation

Compound interest means interest is calculated on the amount including all previously earned interest, not just the original principal. **Depreciation** applies the same logic but as a repeated percentage decrease (e.g. the falling value of a car).

The compound interest formula is:

$$A = P\left(1 + \frac{r}{100}\right)^n$$

where A is the final amount, P is the principal (starting amount), r is the annual interest rate (%), and n is the number of years.

For depreciation, replace the $+$ with a $-$:

$$A = P\left(1 - \frac{r}{100}\right)^n$$

Worked Example:

£5,000 is invested at 3% compound interest per year for 4 years. Find the total amount.

$$A = 5000 \times (1.03)^4 = 5000 \times 1.12550881 = £5{,}627.54 \text{ (to 2 d.p.)}$$

Worked Example:

A car is bought for £18,000. It depreciates at 12% per year. Find its value after 3 years.

$$A = 18{,}000 \times (0.88)^3 = 18{,}000 \times 0.681472 = £12{,}266.50 \text{ (to 2 d.p.)}$$

Worked Example:

How many years does it take for £3,000 invested at 5% compound interest to exceed £4,000?

Try $n = 5$: $3000 \times 1.05^5 = 3000 \times 1.2762... = £3{,}828.84$ (not yet over £4,000)

Try $n = 6$: $3000 \times 1.05^6 = 3000 \times 1.3400... = £4{,}020.19$ (over £4,000)

Answer: 6 years

Simple Interest

Simple interest is calculated only on the original principal each year:

$$\text{Total interest} = \frac{P \times r \times t}{100}$$

where t is the time in years.

Worked Example:

£2,000 is invested at 4% simple interest per year. How much interest is earned over 3 years?

$$\text{Interest} = \frac{2000 \times 4 \times 3}{100} = £240$$

Total amount = £2,000 + £240 = **£2,240**

Feature	Simple Interest	Compound Interest
Interest earned on	Original principal only	Principal + all previous interest
Formula	$I = \frac{Prt}{100}$	$A = P(1 + \frac{r}{100})^n$
Growth pattern	Linear (straight line)	Exponential (curve)
Long-term value	Lower	Higher (grows faster over time)

Exam Tip: For compound interest questions, use your calculator's power function $(1.r)^n$ rather than multiplying repeatedly. This is faster and avoids rounding errors between steps.